Lecture 1: Introduction

The goal of this class is to teach you to **solve** computation problems, and to **communicate** that your solutions are **correct** and **efficient**.

Problem

- Binary relation from **problem inputs** to **correct outputs**
- Usually don't specify every correct output for all inputs (too many!)
- Provide a verifiable **predicate** (a property) that correct outputs must satisfy
- 6.006 studies problems on large general input spaces
- Not general: small input instance
  - **Example:** In this room, is there a pair of students with same birthday?
- General: arbitrarily large inputs
  - **Example:** Given any set of n students, is there a pair of students with same birthday?
  - If birthday is just one of 365, for $n > 365$, answer always true by pigeon-hole
  - Assume resolution of possible birthdays exceeds n (include year, time, etc.)

Algorithm

- Procedure mapping each input to a **single** output (deterministic)
- Algorithm **solves** a problem if it returns a correct output for every problem input
- **Example:** An algorithm to solve birthday matching
  - Maintain a **record** of names and birthdays (initially empty)
  - Interview each student in some order
    - If birthday exists in record, return found pair!
    - Else add name and birthday to record
  - Return None if last student interviewed without success

Correctness

- Programs/algorithms have fixed size, so how to prove correct?
- For small inputs, can use case analysis
- For arbitrarily large inputs, algorithm must be **recursive** or loop in some way
- Must use **induction** (why recursion is such a key concept in computer science)
- **Example:** Proof of correctness of birthday matching algorithm
  - Induct on k: the number of students in record
  - **Hypothesis:** if first k contain match, returns match before interviewing student $k+1$
  - **Base case**: $k = 0$, first k contains no match
  - Assume for induction hypothesis holds for $k = k'$, and consider $k = k' + 1$
  - If first k' contains a match, already returned a match by induction
  - Else first k' do not have match, so if first $k' + 1$ has match, match contains $k' + 1$
  - Then algorithm checks directly whether birthday of student $k' + 1$ exists in first k' $\square$

Efficiency

- How fast does an algorithm produce a correct output?
  - Could measure time, but want performance to be machine independent
  - **Idea!** Count number of fixed-time operations algorithm takes to return
  - Expect to depend on size of input: larger input suggests longer time
  - Size of input is often called 'n', but not always!
  - Efficient if returns in **polynomial time** with respect to input
  - Sometimes no efficient algorithm exists for a problem! (See L20)
- Asymptotic Notation: ignore constant factors and low order terms
  - Upper bounds (O), lower bounds (Ω), tight bounds (Θ) $\in, =$, is, order
  - Time estimate below based on one operation per cycle on a 1 GHz single-core machine
  - Particles in universe estimated $< 10^{100}$

input	constant	logarithmic	linear	log-linear	quadratic	polynomial	exponential
n	$\Theta(1)$	$\Theta(\log n)$	$\Theta(n)$	$\Theta(n \log n)$	$\Theta(n^2)$	$\Theta(n^c)$	$2^{\Theta(n^c)}$
1000	1	≈ 10	1000	$\approx$ 10,000	1,000,000	1000^c	$2^{1000} \approx 10^{301}$
Time	$1\,ns$	$10\,ns$	$1\,\mu s$	$10\,\mu s$	$1\,ms$	$10^{3c-9}\,s$	10^{281} millenia

Model of Computation

- Specification for what operations on the machine can be performed in $O(1)$ time
- Model in this class is called the **Word-RAM**
- **Machine word**: block of w bits (w is word size of a w-bit Word-RAM)
- **Memory**: Addressable sequence of machine words
- **Processor** supports many **constant time** operations on a $O(1)$ number of words (**integers**):
  - **integer** arithmetic: `(+, -, *, //, %)`
  - **logical** operators: `(&&, ||, !, ==, <, >, <=, =>)`
  - (**bitwise** arithmetic: `(&, |, <<, >>, ...)`)
  - Given word a, can **read** word at address a, **write** word to address a
- Memory address must be able to access every place in memory
  - Requirement: $w \geq$ # bits to represent largest memory address, i.e., $\log_2 n$
  - 32-bit words $\rightarrow$ max ~ 4 GB memory, 64-bit words $\rightarrow$ max ~ 16 exabytes of memory
- **Python** is a more complicated model of computation, implemented on a Word-RAM

Data Structure

- A **data structure** is a way to store non-constant data, that supports a set of operations
- A collection of operations is called an **interface**
  - **Sequence:** Extrinsic order to items (first, last, nth)
  - **Set:** Intrinsic order to items (queries based on item keys)
- Data structures may implement the same interface with different performance
- **Example: Static Array** - fixed width slots, fixed length, static sequence interface
  - `StaticArray(n)`: allocate static array of size n initialized to 0 in $\Theta(n)$ time
  - `StaticArray.get_at(i)`: return word stored at array index i in $\Theta(1)$ time
  - `StaticArray.set_at(i, x)`: write word x to array index i in $\Theta(1)$ time
- Stored word can hold the address of a larger object
- Like Python `tuple` plus `set_at(i, x)`, Python `list` is a **dynamic array** (see L02)

```python
def birthday_match(students):
    '''
    Find a pair of students with the same birthday
    Input:  tuple of student (name, bday) tuples
    Output: tuple of student names or None
    '''
    n = len(students)                           # O(1)
    record = StaticArray(n)                     # O(n)
    for k in range(n):                          # n
        (name1, bday1) = students[k]            # O(1)
        # Return pair if bday1 in record
        for i in range(k):                      # k
            (name2, bday2) = record.get_at(i)   # O(1)
            if bday1 == bday2:                  # O(1)
                return (name1, name2)           # O(1)
        record.set_at(k, (name1, bday1))        # O(1)
    return None                                 # O(1)
```

Example: Running Time Analysis

- Two loops: outer $k \in \{0, \ldots, n-1\}$, inner is $i \in \{0, \ldots, k\}$
- Running time is $O(n) + \sum_{k=0}^{n-1}(O(1) + k \cdot O(1)) = O(n^2)$
- Quadratic in n is **polynomial**. Efficient? Use different data structure for record!

How to Solve an Algorithms Problem

1. Reduce to a problem you already know (use data structure or algorithm)

Search Problem (Data Structures)	Sort Algorithms	Shortest Path Algorithms
Static Array (L01)	Insertion Sort (L03)	Breadth First Search (L09)
Linked List (L02)	Selection Sort (L03)	DAG Relaxation (L11)
Dynamic Array (L02)	Merge Sort (L03)	Depth First Search (L10)
Sorted Array (L03)	Counting Sort (L05)	Topological Sort (L10)
Direct-Access Array (L04)	Radix Sort (L05)	Bellman-Ford (L12)
Hash Table (L04)	AVL Sort (L07)	Dijkstra (L13)
Balanced Binary Tree (L06-L07)	Heap Sort (L08)	Johnson (L14)
Binary Heap (L08)		Floyd-Warshall (L18)

2. Design your own (recursive) algorithm
   - Brute Force
   - Decrease and Conquer
   - Divide and Conquer
   - **Dynamic Programming** (L15-L19)
   - Greedy / Incremental

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